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Profile

Experience: Machine Learning Engineer with 7.8 years in NLP, AWS. 4+ years in Generative AI, LLMs, prompt engineering, LangChain, LangGraph.

Achievements: Reduced regulatory document authoring from weeks to days. Delivered multi-agent systems, hierarchical RAG pipelines, and NER models with 90%+ accuracy.

Leadership: Promoted to Senior Software Engineer. Led teams of 5+, managing full project lifecycles and mentoring engineers.

Deployment: Expert in productionizing AI systems using AWS (S3, ECS, SageMaker, OpenSearch) and Azure OpenAI, ensuring scalability in regulated environments.

Tech Stack: Azure OpenAI (GPT-4o), LangGraph, LangChain, RAG, Few-Shot prompting, Python, FastAPI, Docker, Vector Databases, Agentic AI

Availability: Immediate joiner

Professional Experience

Software Engineer II, ValueLabs, Bengaluru

Dec 2022 — Present

Projects

AI-Powered CTD Module 2 (QOS) Generator

- **Problem:** The manual authoring of the Quality Overall Summary (QOS) for regulatory drug submissions is a major industry bottleneck. This process is extremely time-consuming (taking weeks to months), highly susceptible to human error, and creates a significant compliance risk due to potential inconsistencies. Furthermore, the specialized "tribal knowledge" required is often lost through employee turnover.
- **Solution:** Engineered an end-to-end AI platform that automates the generation of the CTD Module 2.5 (QOS). The system ingests source technical data (Module 3) and a corpus of successful historical submissions. A fine-tuned LLM (GPT-4o) then uses a "few-shot learning" approach to generate a draft summary, section-by-section, that matches the company's specific tone and narrative style. The entire workflow is managed through a "human-in-the-loop" web dashboard for expert review and final approval.
- **Benefits:**
  Accelerated Timelines: Reduced the QOS authoring and review cycle from **weeks or months down to days**, directly impacting submission speed.
  Enhanced Compliance: Mitigated regulatory risk by programmatically ensuring consistency between source data and the final summary, minimizing scrutiny from health authorities.
  Increased SME Productivity: Liberated Subject Matter Experts from thousands of hours of manual drafting, allowing them to focus on high-value scientific review and regulatory strategy.
- **Skills and Tools:**
  AI/ML: Azure OpenAI (GPT-4o), LLM Prompt Engineering, Few-Shot Learning, Vector Embeddings
  Backend: Python, FastAPI, python-docx, Docker
  Frontend: React.js, Microsoft Office Add-in
  Cloud & DevOps: AWS (S3, ECS, Lambda), Event-Driven Architecture

Hierarchical RAG Pipeline for Enhanced Contextual Retrieval

- **Problem:** Traditional Retrieval-Augmented Generation (RAG) systems use a fixed-size chunking strategy that breaks the logical structure of documents. This provides the LLM with disjointed, context-poor text fragments, leading to inaccurate, incomplete answers and a failure to grasp the broader narrative.
- **Solution:** Architected a novel hierarchical RAG pipeline that preserves the document's native structure. During ingestion, documents are parsed into a logical tree; full-text parent sections are stored in **AWS S3**, while smaller, granular child chunks are vectorized and indexed in **AWS OpenSearch**. At query time, the system performs a search to identify the most relevant child chunks but then retrieves their complete, context-rich parent sections from S3 to augment the LLM prompt.
- **Benefits:**
  Superior Context: Empowered the LLM with complete, structurally-aware document sections instead of isolated fragments, dramatically improving its ability to understand nuances and relationships.
  Enhanced Accuracy: Significantly increased the accuracy, relevance, and comprehensiveness of the generated answers by effectively solving the "lost-in-the-middle" problem.
  Improved Reliability: Built a more robust and dependable RAG system by ensuring the full logical context is always considered before generating a response.
- **Skills and Tools:**
  Cloud & Storage: AWS (S3, OpenSearch), Azure Document Intelligence
  AI/ML: RAG, Vector Embeddings (Sentence-BERT, OpenAI), LLMs (GPT-4, Claude 3)
  Data Processing: Python, , Hierarchical Data Structures

python-docx

Documender Multi-Agent Enhancement

- **Problem:** The original Documender tool operated as a monolithic script that produced inconsistent output. This required extensive and time-consuming manual review cycles from multiple experts (e.g., legal, medical, regulatory) to ensure the final document met quality and compliance standards, creating a significant workflow bottleneck.
- **Solution:** Refactored the application into a reusable component and architected a sophisticated multi-agent system using LangGraph. The new workflow automates a critique-and-refinement loop where agents generate a multi-point scoring rubric for each requirement, and a agent iteratively improves the document until it achieves a target quality score.

Critique
Writer

- **Benefits:**
  Automated Multi-Persona Review: Introduced the capability to simulate a review from multiple expert personas (e.g., medical writer, legal counsel) within the generation process, ensuring the output is compliant and well-rounded from the first draft.
  Drastically Reduced Review Cycles: By embedding an automated, rubric-based quality gate, the system significantly reduced the need for lengthy manual

reviews by human experts, accelerating content finalization.
Improved Output Quality & Consistency: The iterative refinement process against a scoring rubric enforced a consistent, high standard of quality for all generated documents.

- **Skills and Tools:**
AI/ML: LangGraph, GPT-4o, Multi-Agent Systems
Backend & DevOps: Python, FastAPI, Docker

CTD Colour Extractor Utility (Formatted)

- **Problem:** Manually creating distinct, region-specific versions of regulatory documents (CTD Module 3) from a core document was a major bottleneck, requiring an estimated **150 hours of skilled labor** per molecule. This process was highly error-prone and created significant challenges for version control.
- **Solution:** Developed an automation tool based on a "single source of truth" methodology. A writer creates one master Word document with region-specific text color-coded. The tool, via a Vue.js front-end, ingests this document, uses **GPT-4o** to interpret the color key, and programmatically generates a complete set of clean, region-specific submission documents using the library, preserving all complex formatting.

python-docx

- **Benefits:**
Massive Efficiency Gain: Slashed document processing time by over **99%**, from **150 hours down to approximately 2 minutes** per molecule.
Enhanced Compliance & Quality: Eliminated manual copy-paste errors and ensured perfect consistency across regional dossiers.
Improved Version Control: Radically simplified updates and maintenance by centralizing all content into a single master document.
- **Skills and Tools:** Python, , GPT-4o, Vue.js, Document Automation.

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Software Engineer, Amadeus, Bangalore

Nov 2021 — Dec 2022

Redaction Bot

- **Problem**: The publishing or sharing of numerous clinical and research documents across teams presented a challenge due to the presence of Personal Identification Information (PII) within them. The necessity to redact PII in compliance with regulations such as HIPPA, GDPR, and Health Canada from documents, which could span up to 3,000 pages, placed significant time demands on the writers' team.
- **Solution**: Developed an application empowering writers to forward documents to an RPA bot via email. Upon receipt, this bot initiated a Python-based procedure to identify and redact PII from PDFs. To spot PII data, AWS Comprehend and Hugging Face's NER pretrained BERT Models were employed. The actual masking of PII in PDFs was facilitated by AWS Textract and the Python Fitz library. Post-redaction, the Azure OpenAI Large Language Model Chat GPT-4 was leveraged to validate the Python bot's redaction performance.
- **Benefits**: By streamlining the redaction process and enhancing quality control checks, the application significantly expedited the time required by the writers' team to mask PII within PDF documents.
- **Skills and Tools**: PyMuPDF, Fitz, AWS Textract, AWS Sagemaker, HuggingFace, Langchain, Azure OpenAI, GPT-4, Chain Of Thought Prompting Technique.

Automated Abbreviation Web Portal

- **Problem**: Medical writers were spending several hours manually compiling lists of abbreviations and definitions from clinical documents.
- **Solution**: Developed an automated web portal enabling users to upload clinical documents and download abbreviation lists as Word files. This portal harnesses a deep learning NER model, trained on a dataset of 100,000+ clinical documents, to extract abbreviations and definitions.
- **Benefits**: The portal significantly reduces the time medical writers spend compiling abbreviation lists and enhances accuracy through the deep learning model's precision.
- **Skills and Tools**: Hugging Face, AWS, Python, Pandas, Docx, Named Entity Recognition(NER)

Software Engineer, ValueLabs, Bangalore

Dec 2017 — Nov 2021

Built and deployed a text classification API for Ehelpline

- Built a text classification API to classify helpline tickets that come to Wipro Ehelpline, an internal product used by Valuelabs like Service now.
- Moved the previous API from Elastic Search to Machine Learning, which significantly improved the accuracy of the API.
- Improved the text classification model performance using pre-trained word embeddings and RUN up to 90%.
- Deployed the model as a Flask API for the Ehelpline product.

Projects:

Classifier Bot API for Ehelpline Tool

Problem: Ehelpline is an internal product used by ValueLabs to manage helpline tickets. The previous API for classifying helpline tickets was based on Elastic Search, but it was not very accurate.

Solution: I built a new text classification API using Machine Learning. This API is significantly more accurate than the previous one, and it can be used to classify helpline tickets with up to 90% accuracy.

Benefits: The new API has several benefits, including:

- **Increased accuracy**: The new API is significantly more accurate than the previous one, which means that helpline tickets can be classified more accurately.
- **Reduced manual effort**: The new API can automatically classify helpline tickets, which reduces the amount of manual effort required to manage the helpline.
- **Improved customer experience**: The new API can automatically route helpline tickets to the correct department, which improves the customer experience.

Skills and Tools: Neural Networks, Keras, Tensorflow, Scikit-Learn, Pandas, Flask

Prediction of loan Closing Date (Gallagher Solutions)

Problem: Gallagher Solutions is a financial services company that provides loans to businesses and individuals. The company wants to be able to predict the closing date of a loan application so that they can better manage their resources.

Solution: I developed a machine learning model to predict the closing date of a loan application. The model was trained on a dataset of historical loan applications. The model was able to reduce the mean absolute error (MAE) from 24 to 10

Benefits: The benefits of model include:

- Improved efficiency: The company can now better manage their resources by knowing when to expect loan applications to close.
- Increased accuracy: The company can now make more accurate decisions about loan applications.
- Reduced risk: The company can now reduce the risk of loan defaults by identifying loans that are more likely to close on time.

Skills and Tools: Machine Learning, Data Science, Python, Scikit-Learn, Flask

Education

Btech, Sree Vidyanikethan Engineering college, Tirupati, Bangalore
May 2013 — May 2017
Graduated with CGPA of 8.9

Technical Skills

Python:
Machine Learning:
Deep Learning:
Natural Language Processing:
Generative AI:
Langchain :
Langgraph:

Certifications

Tensorflow Developer Certificate by Tensorflow Google

DeepLearning.AI TensorFlow Developer Specialization Coursera

Neural Networks and Deep Learning Coursera

Introduction To Langgraph

Deepsearch with Langgraph